CS4308 Graph Machine Learning

Fifty papers to read and build

A curated path from graph structure and random walks to neural message passing, transformers, generation, and reliable evaluation. This is a foundational research syllabus, not an exhaustive survey of the newest literature. Bibliographic sources were checked in November 2026. The year shown is the cited publication year; an arXiv preprint may be earlier. Workshop papers are identified explicitly.

How to use this list

The class collectively discusses all 50 papers. Each group, normally a pair, must understand, implement, evaluate, and present at least five distinct papers. Register a balanced selection: one random-walk or spectral method, one message-passing or attention method, one advanced method, one evaluation or reliability paper, and one project-aligned paper. For theory and benchmark papers, implement a central diagnostic or evaluation protocol.

Submit a critical brief, reproducible code, a central result, one ablation or stress test, and a presentation for each study. Small-scale reproductions are acceptable when explicitly scoped. Both partners must contribute and defend the work. Suggested checkpoints are weeks 4, 6, 9, 11, and 13. This reading companion uses the same P01–P50 identifiers as the course outline and website.

GAN means generative adversarial network; GAT means graph attention network. P43 supplies the general GAN prerequisite, while P44 and P45 apply adversarial objectives to graph representation or generation.

Weekly reading map

Week 2: Spectral structure and random walks. P01, P02

Week 3: Learning with walks. P03, P04, P05, P06

Week 4: From graph signals to GCNs. P07, P08, P09

Week 5: Message passing and attention. P10, P11, P12, P15

Week 6: Expressivity and graph prediction. P13, P14, P16, P17, P21

Week 7: When graph learning fails. P18, P19, P20, P22, P23, P24

Week 9: Scaling and self-supervision. P25, P26, P27, P29, P30, P31, P32

Week 10: Relations and time. P33, P34, P35, P36, P37

Week 11: Graph transformers. P38, P39, P40, P41, P42

Week 12: Adversarial and generative graph learning. P43, P44, P45, P46, P47

Week 13: Trustworthy graph research. P28, P48, P49, P50

Foundations and embeddings

P01 The PageRank Citation Ranking: Bringing Order to the Web

Lawrence Page et al. Stanford InfoLab Technical Report 1999-66, 1999.
Why read it. Connects graph centrality, Markov chains, and ranking before learned representations.
Build or investigate. Implement power iteration with teleportation; test dangling nodes and sensitivity to the damping factor.
Level. Foundation Primary source ↗

P02 Laplacian Eigenmaps and Spectral Techniques for Embedding and Clustering

Mikhail Belkin and Partha Niyogi. NeurIPS (then NIPS), 2001.
Why read it. Introduces the graph Laplacian as a bridge between geometry, embeddings, and spectral learning.
Build or investigate. Compute two-dimensional spectral embeddings of a small graph; compare neighborhoods with a random layout.
Level. Foundation Primary source ↗

P03 DeepWalk: Online Learning of Social Representations

Bryan Perozzi et al. KDD, 2014.
Why read it. Turns short random walks into a practical unsupervised representation-learning pipeline.
Build or investigate. Implement walk generation and skip-gram training; evaluate node classification with a fixed split.
Level. Foundation Primary source ↗

P04 LINE: Large-scale Information Network Embedding

Jian Tang et al. WWW, 2015.
Why read it. Separates first-order and second-order proximity and introduces efficient edge sampling.
Build or investigate. Implement the two proximity objectives and compare their concatenation on link prediction.
Level. Foundation Primary source ↗

P05 node2vec: Scalable Feature Learning for Networks

Aditya Grover and Jure Leskovec. KDD, 2016.
Why read it. Shows how biased walks trade off local neighborhoods and structural roles.
Build or investigate. Sweep return and exploration parameters against DeepWalk using identical downstream evaluation.
Level. Foundation Primary source ↗

P06 Network Embedding as Matrix Factorization: Unifying DeepWalk, LINE, PTE, and node2vec

Jiezhong Qiu et al. WSDM, 2018.
Why read it. Explains random-walk embeddings through matrix factorization and spectral graph structure.
Build or investigate. Construct NetMF on a small graph and compare it with sampled DeepWalk embeddings.
Level. Advanced Primary source ↗

Graph neural networks

P07 Spectral Networks and Locally Connected Networks on Graphs

Joan Bruna et al. ICLR, 2014.
Why read it. Provides a historical starting point for learned convolution in a graph Fourier basis.
Build or investigate. Implement one spectral filtering layer and measure how its parameters depend on the graph basis.
Level. Advanced Primary source ↗

P08 Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering

Michaël Defferrard et al. NeurIPS (then NIPS), 2016.
Why read it. Replaces expensive eigendecomposition with localized polynomial filters.
Build or investigate. Implement Chebyshev recurrence; compare polynomial order, receptive field, runtime, and accuracy.
Level. Intermediate Primary source ↗

P09 Semi-Supervised Classification with Graph Convolutional Networks

Thomas N. Kipf and Max Welling. ICLR, 2017.
Why read it. Establishes a compact, reproducible baseline for learning from node features and edges.
Build or investigate. Implement a two-layer GCN; ablate self-loops, normalization, and feature-only baselines.
Level. Foundation Primary source ↗

P10 Inductive Representation Learning on Large Graphs

William L. Hamilton et al. NeurIPS (then NIPS), 2017.
Why read it. Introduces neighborhood sampling and aggregation for generalization to unseen nodes.
Build or investigate. Implement mean aggregation with sampling and evaluate a genuine inductive node split.
Level. Intermediate Primary source ↗

P11 Neural Message Passing for Quantum Chemistry

Justin Gilmer et al. ICML, 2017.
Why read it. Unifies many graph architectures through message, update, and readout functions.
Build or investigate. Build a small edge-aware message-passing model for one molecular regression target.
Level. Intermediate Primary source ↗

P12 Graph Attention Networks

Petar Veličković et al. ICLR, 2018.
Why read it. Introduces learned neighborhood weighting and multi-head attention.
Build or investigate. Implement masked graph attention; compare heads and attention dropout with GCN.
Level. Intermediate Primary source ↗

P13 How Powerful are Graph Neural Networks?

Keyulu Xu et al. ICLR, 2019.
Why read it. Relates aggregation to graph isomorphism and motivates the Graph Isomorphism Network.
Build or investigate. Implement GIN and test sum versus mean aggregation on controlled multiset and graph examples.
Level. Advanced Primary source ↗

P14 Hierarchical Graph Representation Learning with Differentiable Pooling

Rex Ying et al. NeurIPS, 2018.
Why read it. Introduces learned graph coarsening for prediction over entire graphs.
Build or investigate. Implement one DiffPool stage and compare it with global mean pooling under matched budgets.
Level. Intermediate Primary source ↗

P15 Gated Graph Sequence Neural Networks

Yujia Li et al. ICLR, 2016.
Why read it. Connects recurrent gated updates to reasoning on graph structure.
Build or investigate. Implement gated propagation on a synthetic reachability or shortest-path task.
Level. Intermediate Primary source ↗

P16 Simplifying Graph Convolutional Networks

Felix Wu et al. ICML, 2019.
Why read it. Tests whether nonlinear depth is necessary and supplies a strong inexpensive baseline.
Build or investigate. Reproduce SGC and compare propagation depth, training time, and accuracy against GCN.
Level. Foundation Primary source ↗

P17 Predict then Propagate: Graph Neural Networks meet Personalized PageRank

Johannes Gasteiger et al. ICLR, 2019.
Why read it. Connects the opening PageRank material to decoupled neural prediction and diffusion.
Build or investigate. Implement APPNP and sweep teleportation probability and propagation steps.
Level. Intermediate Primary source ↗

Expressivity and limitations

P18 Geom-GCN: Geometric Graph Convolutional Networks

Hongbin Pei et al. ICLR, 2020.
Why read it. Explores geometric neighborhoods when ordinary local aggregation misses useful structure.
Build or investigate. Compare geometric and adjacency neighborhoods using documented public data splits.
Level. Advanced Primary source ↗

P19 Beyond Homophily in Graph Neural Networks: Current Limitations and Effective Designs

Jiong Zhu et al. NeurIPS, 2020.
Why read it. Studies graphs whose connected nodes differ and motivates separating ego and neighbor information.
Build or investigate. Implement an H2GCN-style baseline and vary homophily on synthetic graphs.
Level. Intermediate Primary source ↗

P21 Weisfeiler and Leman Go Neural: Higher-order Graph Neural Networks

Christopher Morris et al. AAAI, 2019.
Why read it. Examines the expressivity ceiling of standard message passing and higher-order alternatives.
Build or investigate. Construct graph pairs that defeat 1-WL and compare a small higher-order model.
Level. Advanced Primary source ↗

P22 On the Bottleneck of Graph Neural Networks and its Practical Implications

Uri Alon and Eran Yahav. ICLR, 2021.
Why read it. Introduces over-squashing as a failure of information transmission across graph bottlenecks.
Build or investigate. Reproduce a small long-range task and compare added shortcuts with increased hidden dimension.
Level. Advanced Primary source ↗

P23 Understanding over-squashing and bottlenecks on graphs via curvature

Jake Topping et al. ICLR, 2022.
Why read it. Gives a geometric account of bottlenecks and a principled graph-rewiring approach.
Build or investigate. Implement or adapt curvature-based rewiring; compare against random edge additions at equal budget.
Level. Advanced Primary source ↗

P24 PairNorm: Tackling Oversmoothing in GNNs

Lingxiao Zhao and Leman Akoglu. ICLR, 2020.
Why read it. Separates representation collapse with depth from other GNN limitations.
Build or investigate. Implement PairNorm and plot node-embedding distances and accuracy as depth increases.
Level. Intermediate Primary source ↗

Evaluation and trust

P20 Pitfalls of Graph Neural Network Evaluation

Oleksandr Shchur et al. NeurIPS Relational Representation Learning Workshop, 2018.
Why read it. Makes split choice, fair tuning, and uncertainty central to claims about model quality.
Build or investigate. Implement a shared evaluation harness for GCN and GAT across several splits and random seeds.
Level. Foundation Primary source ↗

P28 Open Graph Benchmark: Datasets for Machine Learning on Graphs

Weihua Hu et al. NeurIPS, 2020.
Why read it. Supplies task-specific datasets, realistic splits, and standardized evaluators.
Build or investigate. Implement an OGB baseline and document why its official split and metric match the task.
Level. Foundation Primary source ↗

P48 GNNExplainer: Generating Explanations for Graph Neural Networks

Rex Ying et al. NeurIPS, 2019.
Why read it. Explores which subgraphs and features support a GNN prediction.
Build or investigate. Implement explanation masks on a motif dataset; measure fidelity, sparsity, and stability.
Level. Intermediate Primary source ↗

P49 Adversarial Attacks on Graph Neural Networks via Meta Learning

Daniel Zügner and Stephan Günnemann. ICLR, 2019.
Why read it. Studies sensitivity to training-graph perturbations and motivates robust evaluation.
Build or investigate. On a public benchmark, compare controlled edge perturbations with random edits and report accuracy degradation.
Level. Advanced Primary source ↗

P50 Temporal Graph Benchmark for Machine Learning on Temporal Graphs

Shenyang Huang et al. NeurIPS Datasets and Benchmarks Track, 2023.
Why read it. Makes temporal splits, realistic negative candidates, and simple baselines part of credible research.
Build or investigate. Implement an official temporal baseline and compare it with a recent-history heuristic.
Level. Intermediate Primary source ↗

Scalable learning

P25 Cluster-GCN: An Efficient Algorithm for Training Deep and Large Graph Convolutional Networks

Wei-Lin Chiang et al. KDD, 2019.
Why read it. Uses graph partitions to make minibatch training memory-efficient.
Build or investigate. Train on cluster batches; report peak memory, epoch time, and accuracy against neighbor sampling.
Level. Intermediate Primary source ↗

P26 GraphSAINT: Graph Sampling Based Inductive Learning Method

Hanqing Zeng et al. ICLR, 2020.
Why read it. Studies subgraph sampling with normalization to improve training efficiency.
Build or investigate. Implement one subgraph sampler and test the contribution of sampling normalization.
Level. Intermediate Primary source ↗

P27 SIGN: Scalable Inception Graph Neural Networks

Fabrizio Frasca et al. ICML Graph Representation Learning and Beyond Workshop, 2020.
Why read it. Shows how precomputed graph filters can trade storage for fast training.
Build or investigate. Precompute multiple propagation scales and include preprocessing time in comparisons with GraphSAGE.
Level. Intermediate Primary source ↗

Self-supervised learning

P29 Deep Graph Infomax
Petar Veličković et al. ICLR, 2019.
Why read it. Introduces node representation learning through local-global contrast.
Build or investigate. Implement DGI and test feature corruption with a frozen linear evaluation protocol.
Level. Intermediate Primary source ↗

P30 Graph Contrastive Learning with Augmentations
Yuning You et al. NeurIPS, 2020.
Why read it. Makes graph augmentations a central design choice in contrastive pretraining.
Build or investigate. Implement two graph augmentations and assess whether they preserve labels on the chosen dataset.
Level. Intermediate Primary source ↗

P31 Large-Scale Representation Learning on Graphs via Bootstrapping
Shantanu Thakoor et al. ICLR, 2022.
Why read it. Explores self-supervision without explicit negative pairs and its memory implications.
Build or investigate. Implement a teacher-student moving-average update and ablate stop-gradient or predictor design.
Level. Advanced Primary source ↗

P32 GraphMAE: Self-Supervised Masked Graph Autoencoders
Zhenyu Hou et al. KDD, 2022.
Why read it. Uses masked feature reconstruction as an alternative to contrastive objectives.
Build or investigate. Implement masked reconstruction and compare masking ratios with a supervised baseline.
Level. Intermediate Primary source ↗

Heterogeneous and temporal graphs

P33 Modeling Relational Data with Graph Convolutional Networks
Michael Schlichtkrull et al. ESWC, 2018.
Why read it. Extends message passing to typed relations and knowledge-graph completion.
Build or investigate. Implement basis-decomposed R-GCN on a small relational dataset with leakage-aware edge splits.
Level. Intermediate Primary source ↗

P34 Heterogeneous Graph Attention Network
Xiao Wang et al. WWW, 2019.
Why read it. Separates attention over neighbors from attention over semantic meta-paths.
Build or investigate. Implement two meta-path channels and compare learned semantic attention with uniform weighting.
Level. Intermediate Primary source ↗

P35 Heterogeneous Graph Transformer
Ziniu Hu et al. WWW, 2020.
Why read it. Combines type-dependent attention, temporal encodings, and heterogeneous sampling.
Build or investigate. Implement type-specific projections and ablate node and relation types on a manageable graph.
Level. Advanced Primary source ↗

P36 Inductive Representation Learning on Temporal Graphs
Da Xu et al. ICLR, 2020.
Why read it. Introduces temporal attention with continuous-time encodings.
Build or investigate. Implement TGAT-style temporal neighbors and verify every prediction uses only past events.
Level. Advanced Primary source ↗

P37 Temporal Graph Networks for Deep Learning on Dynamic Graphs
Emanuele Rossi et al. ICML Graph Representation Learning and Beyond Workshop, 2020.
Why read it. Introduces event-driven memory for learning on changing graphs.
Build or investigate. Implement a reduced TGN pipeline and compare memory updates with a memory-free temporal model.
Level. Advanced Primary source ↗

Graph transformers

P38 A Generalization of Transformer Networks to Graphs
Vijay Prakash Dwivedi and Xavier Bresson. AAAI Workshop on Deep Learning on Graphs, 2021.
Why read it. Introduces a clear graph-transformer baseline with Laplacian positional information.
Build or investigate. Implement sparse graph attention with Laplacian encodings and test the effect of removing positions.
Level. Intermediate Primary source ↗

P39 Do Transformers Really Perform Bad for Graph Representation?
Chengxuan Ying et al. NeurIPS, 2021.
Why read it. Shows how structural encodings make global attention useful for graph prediction.
Build or investigate. Implement a compact Graphormer and ablate degree and shortest-path encodings.
Level. Advanced Primary source ↗

P40 Rethinking Graph Transformers with Spectral Attention
Devin Kreuzer et al. NeurIPS, 2021.
Why read it. Studies learned spectral position encodings and fully connected attention.
Build or investigate. Implement a small spectral encoder and examine eigenvector-sign perturbations and graph-size cost.
Level. Advanced Primary source ↗

P41 Recipe for a General, Powerful, Scalable Graph Transformer
Ladislav Rampášek et al. NeurIPS, 2022.
Why read it. Combines local message passing, global attention, and position or structure encodings.
Build or investigate. Implement one GPS block and perform a three-component ablation under a fixed parameter budget.
Level. Advanced Primary source ↗

P42 Exphormer: Sparse Transformers for Graphs
Hamed Shirzad et al. ICML, 2023.
Why read it. Uses sparse attention patterns to make graph transformers practical on larger graphs.
Build or investigate. Implement expander-style sparse attention and compare memory and accuracy with dense attention.
Level. Advanced Primary source ↗

Generative graph learning

P43 Generative Adversarial Nets
Ian J. Goodfellow et al. NeurIPS (then NIPS), 2014.
Why read it. Provides the adversarial-learning prerequisite; this paper itself is not a graph model.
Build or investigate. Implement a small GAN on a toy distribution and diagnose instability before attempting graph generation.
Level. Foundation Primary source ↗

P44 GraphGAN: Graph Representation Learning with Generative Adversarial Nets
Hongwei Wang et al. AAAI, 2018.
Why read it. Applies adversarial training to connectivity distributions for graph embeddings.
Build or investigate. Implement a small generator-discriminator link model and compare with a random-walk embedding baseline.
Level. Advanced Primary source ↗

P45 MolGAN: An implicit generative model for small molecular graphs
Nicola De Cao and Thomas Kipf. ICML Workshop on Theoretical Foundations and Applications of Deep Generative Models, 2018.
Why read it. Combines adversarial graph generation with property objectives and exposes mode collapse.
Build or investigate. Implement a reduced small-molecule generator; report validity, uniqueness, novelty, and diversity.
Level. Advanced Primary source ↗

P46 GraphRNN: Generating Realistic Graphs with Deep Auto-regressive Models
Jiaxuan You et al. ICML, 2018.
Why read it. Provides an autoregressive alternative to adversarial graph generation.
Build or investigate. Implement a small sequential graph generator and compare degree and clustering distributions.
Level. Advanced Primary source ↗

P47 DiGress: Discrete Denoising Diffusion for Graph Generation
Clément Vignac et al. ICLR, 2023.
Why read it. Introduces categorical diffusion for graph structure and attributes.
Build or investigate. Implement a reduced discrete diffusion model on small synthetic graphs and evaluate structural validity.
Level. Advanced Primary source ↗

Twenty illustrative research projects

A rigorous replication with a meaningful extension, a careful negative result, or a focused new method can all make a strong project. Publication acceptance is not a grading criterion. The required paper is submitted to the course; external conference or journal submission is optional and requires agreement among the authors.

These are scoped research suggestions. Agree on a dataset, question, and compute budget before beginning. Each project should report competitive and simple baselines, valid splits, ablations, error analysis, and reproducible artifacts.

R01 When do biased walks help?

Which graph structures make node2vec's walk bias useful?
Data. Synthetic stochastic block models and Cora. Dataset documentation
Baselines. DeepWalk, node2vec, degree features plus logistic regression
Evaluation. Node macro-F1 and accuracy; sweep homophily, walk length, p and q under fixed embedding budgets.
Extension. Relate performance to mixing time and community strength.
Compute. CPU-friendly

R02 PageRank as a learning prior

Can personalized diffusion improve classification when labels are scarce?
Data. Cora and CiteSeer. Dataset documentation
Baselines. MLP, GCN, personalized PageRank propagation
Evaluation. Accuracy versus labeled fraction; fix splits and compare propagation depth and damping.
Extension. Choose damping using local graph structure without test labels.
Compute. CPU or small GPU

R03 Attention under noisy edges

Does learned graph attention remain useful as irrelevant edges increase?
Data. Cora plus controlled random edge additions. Dataset documentation
Baselines. GCN, GraphSAGE, GAT
Evaluation. Accuracy and calibration versus edge corruption; match parameter counts and tuning budgets.
Extension. Compare attention entropy with robustness and test a sparsity regularizer.
Compute. Small GPU

R04 Learning across unlike neighbors

When do neighborhood aggregation models fail on heterophilous graphs?
Data. Roman-empire and Minesweeper benchmark splits. Dataset documentation
Baselines. MLP, GCN, GAT and a heterophily-aware model
Evaluation. Dataset-appropriate accuracy or ROC-AUC; preserve supplied splits and report multiple seeds.
Extension. Adapt the balance of self features and neighbor features.
Compute. Small GPU

R05 How deep can a GNN go?

Which changes preserve predictive information as network depth increases?
Data. Cora and synthetic community graphs. Dataset documentation
Baselines. Plain GCN, residual GCN and normalization variants
Evaluation. Accuracy, embedding similarity and Dirichlet energy across depth; control training budget.
Extension. Separate optimization failure from oversmoothing with controlled interventions.
Compute. CPU or small GPU

R06 Rewiring graph bottlenecks

Can a few added edges improve long-range information transfer?
Data. Synthetic trees, barbell graphs and distance-label tasks. Dataset documentation
Baselines. GCN or MPNN with no rewiring, random rewiring and a topology-aware rule
Evaluation. Long-range task accuracy, path length and runtime at equal edge budgets.
Extension. Study the trade-off between shortcut benefits and structural distortion.
Compute. CPU-friendly

R07 A fair transformer comparison

When does global attention earn its extra compute?
Data. ZINC benchmark graphs. Dataset documentation
Baselines. GIN, a graph transformer and a local/global hybrid
Evaluation. Regression MAE, peak memory and runtime; match parameters and training budget.
Extension. Identify graph sizes or structures where global attention helps.
Compute. Small GPU; use a declared subset for pilots

R08 Which positional encoding matters?

How stable are graph transformers to positional encoding choices?
Data. ZINC and synthetic symmetric graphs. Dataset documentation
Baselines. Transformer without positional encoding, Laplacian encoding and random-walk encoding
Evaluation. MAE and permutation consistency; vary signs, eigenspaces and graph size.
Extension. Test sign-invariant or noise-robust encoding variants.
Compute. Small GPU

R09 Self-supervision with few labels

Which graph augmentations help when supervision is limited?
Data. Cora and CiteSeer. Dataset documentation
Baselines. Supervised GCN, DGI and a contrastive graph encoder
Evaluation. Linear-probe accuracy across label budgets; tune augmentations on validation data only.
Extension. Replace uniform edge dropping with a structure-aware rule.
Compute. Small GPU

R10 Masked graph reconstruction

Can reconstructing masked features improve robustness to missing attributes?
Data. Cora and PubMed. Dataset documentation
Baselines. MLP, GCN and GraphMAE-style pretraining
Evaluation. Classification accuracy and reconstruction loss across random and structured missingness.
Extension. Compare degree-aware masking with uniform masking.
Compute. Small GPU

R11 Predicting future collaborations

How much do graph embeddings improve over simple link heuristics?
Data. ogbl-collab. Dataset documentation
Baselines. Common neighbors, Adamic-Adar, node2vec and GraphSAGE
Evaluation. Official Hits@50 with official temporal splits and negative candidates; document allowed edges.
Extension. Add recency-aware edge weights and audit temporal leakage.
Compute. Moderate GPU; prototype on a declared subset

R12 Learning from event streams

When does temporal memory beat a static graph encoder?
Data. Wikipedia temporal interaction data distributed with TGN. Dataset documentation
Baselines. Static GraphSAGE, a recency heuristic and TGN
Evaluation. Average precision and ROC-AUC with time-ordered splits and explicit negative sampling.
Extension. Test memory resets, time encodings and unseen-node performance.
Compute. Small GPU

R13 Relations in a knowledge graph

Do relation-aware messages improve completion beyond embedding models?
Data. FB15k-237. Dataset documentation
Baselines. DistMult, TransE and R-GCN with a common scoring head where appropriate
Evaluation. Filtered MRR and Hits@10; prevent inverse-relation leakage and report relation-wise errors.
Extension. Allocate relation parameters by frequency or sparsity.
Compute. Small to moderate GPU

R14 Graph recommendations beyond popularity

Can graph propagation improve recommendation for low-activity users?
Data. MovieLens 100K stable benchmark. Dataset documentation
Baselines. Popularity, matrix factorization and LightGCN
Evaluation. Recall@20 and NDCG@20 on a time-ordered split; keep candidate sampling fixed.
Extension. Analyze accuracy and coverage across user-activity groups without inferring sensitive attributes.
Compute. CPU or small GPU

R15 Molecular learning under distribution shift

Do stronger graph encoders generalize to new molecular scaffolds?
Data. ogbg-molhiv. Dataset documentation
Baselines. Fingerprint classifier, GCN and GIN
Evaluation. Official ROC-AUC on scaffold splits; add calibration and performance by scaffold frequency.
Extension. Compare pretraining or edge-feature treatments under equal tuning budgets.
Compute. Small GPU

R16 Generating diverse small graphs

Which generative model best preserves structure without memorization?
Data. Synthetic community and grid graphs. Dataset documentation
Baselines. Erdos-Renyi or block-model generator, GraphRNN and a small graph GAN
Evaluation. Validity, uniqueness, novelty, and degree/clustering distribution distances on held-out graphs.
Extension. Diagnose mode collapse as graph size and training diversity change.
Compute. Small GPU

R17 Adversarial versus diffusion generation

How do adversarial and diffusion objectives trade quality for cost?
Data. Small synthetic graphs; optional QM9 extension. Dataset documentation
Baselines. A MolGAN-inspired discrete generator and a small DiGress-style model
Evaluation. Validity, diversity, novelty, structural fidelity and training/sample time at matched graph sizes.
Extension. If using molecules, add chemical-validity checks and compare conditional generation.
Compute. Moderate GPU; synthetic scope recommended

R18 Are graph explanations stable?

Does a model explanation persist across seeds and harmless perturbations?
Data. Synthetic motif graphs with known generating motifs. Dataset documentation
Baselines. Gradient saliency, random edge masks and GNNExplainer
Evaluation. Fidelity, sparsity, motif precision/recall and agreement across repeated runs.
Extension. Build a stability-aware explanation selection rule; distinguish explanation from causality.
Compute. CPU or small GPU

R19 Robustness on a fixed edge budget

Which defenses withstand targeted structural perturbations?
Data. Cora or CiteSeer public benchmark graphs. Dataset documentation
Baselines. GCN, GAT, edge-drop training and a simple graph denoiser
Evaluation. Clean and perturbed accuracy across equal budgets; distinguish random from targeted perturbations.
Extension. Test whether defense gains persist on unseen perturbation strategies in an offline benchmark.
Compute. Small GPU

R20 Do benchmark rankings survive protocol changes?

How sensitive are model rankings to splits, tuning and random seeds?
Data. Cora plus ogbn-arxiv, with a smaller scoped pilot if needed. Dataset documentation
Baselines. MLP, GCN, GraphSAGE and GAT
Evaluation. Accuracy distributions, rank stability, runtime and tuning cost; preserve official test protocols.
Extension. Measure how small search budgets alter conclusions and publish a reproducibility checklist.
Compute. Small to moderate GPU